LiDAR-Guided Reconstruction of Axial Motion Blur with Coded Monitor Illumination

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Abstract—We developed a camera system to study blur as a printed target moves towards the camera during a single exposure. We recorded the target's depth during the hand movement using a static depth sensor, while a monitor modulated the light. We acquired twelve independent pushes under steady-white and coded illumination with two motion instructions. The main comparison uses six coded exposures and changes only the depth history used for reconstruction. Measured depth gives a mean peak signal-to-noise ratio gain of 3.04 decibels over the endpoint-linear model, with a sample standard deviation of 3.67 decibels. Improvement occurred in five of six trials. The mean of the structural-similarity gain is 0.084. In five of the six trials the improvement also occurred for comparison with the blurred input. A different trial did not pass that comparison. Stationary controls, timing fits and sensitivity tests support a conditional advantage from measured motion. Reference-assisted recovery is used, and optical timing, target pose and focus still limit recovery.

Index Terms—Axial motion, LiDAR, coded illumination, image reconstruction, experimental characterization.

1 Introduction

Targets that approach a camera will appear to get larger in the image across the duration of an exposure. Image points will be blurred by varying amounts so a translation-invariant convolution cannot generally describe it. A hand push will also vary in speed. A measurement of depth during exposure might be better able to estimate this evolving magnification. A change in coded illumination changes the contribution from every position of the target in a photograph.

We ask whether measured depth during exposure can reconstruct the same coded image better than a straight line between the same start and end depths. We keep the image, history of illumination, geometry of the endpoints, solver, and the region in which we scored the result, the same for the comparison. We also compare each reconstructed image with the input blur, to see whether a better estimation of motion really can improve the image.

We used a manually-created guide path and an existing upright target stand. We used the camera, depth board and mounting plate from the lab kit. Monitor illumination and the concurrent image, depth, and command logs enable us to model motion during a conventional long exposure. We align these records, build a scale-blur operator, and use records to estimate the sharp image. All motion was produced by hand. We present here the supplementary physical dataset and all of the primary outcomes - the previous five-trial pilot used a different protocol and we keep it separate.

2 Literature Review

Raskar et al. applied temporal coding using a fluttered shutter to encode motion within the exposure and deblur images [1]. We draw inspiration for the use of temporal coding, implementing the method using illumination and modeling axial scaling of the image. Our binary pattern is not optimized and is quite short. Their lateral-motion results are motivating to conduct the experiment, but do not constitute a demonstration that coding will benefit this apparatus.

We apply a total variation (TV) regularization to keep noise down, while still preserving edges [2]. The solver is an accelerated proximal-gradient algorithm based on FISTA [3]. It involves an approximate TV proximal step, for which we check numerical stability. Endpoint registration is based on scale-invariant features [4]. Pixel error is measured with peak signal-to-noise ratio (PSNR), and the quality of local structure is measured by structural similarity (SSIM) [5]. The inclusion of the blurred input image as a baseline provides a check of whether the resulting reconstruction helps to make a better photo in addition to the assumed motion model.

3 Method and Experimental Apparatus

3.1 Physical implementation

Tripod mounted are the OV5647 camera, the LightRanger 14 Click (TMF8829), and the Raspberry Pi 3 B+. The nominal focal length of the lens used is about 30 mm, and the Lab 1 kit lens was retained. The depth board is taped above the camera, leaving the depth sensor's optical window clear. Two team-built guide edges constrain a sliding base on which the printed USAF target sits in an existing upright stand (Fig. 1). The paper is lit on the side by an ASUS ROG Strix XG27UCGR monitor (27-inch, native resolution 3840×2160). We verified that the chosen depth zones viewed the paper since the monitor position had previously been placed in the ranging field resulting in foreground return signals.

The base front part is moved from 75 to 65 cm on the target axis toward the camera. The paper is located 22 mm past the front edge, so the distance $D_{\mathrm{paper}} = D_{\mathrm{edge}} + 22\,\mathrm{mm}$. This position-reference conversion does not measure sensor bias, as the ruler zero was not surveyed against the optical origin.

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a Manual target path and monitor



b Fixed camera and depth sensor

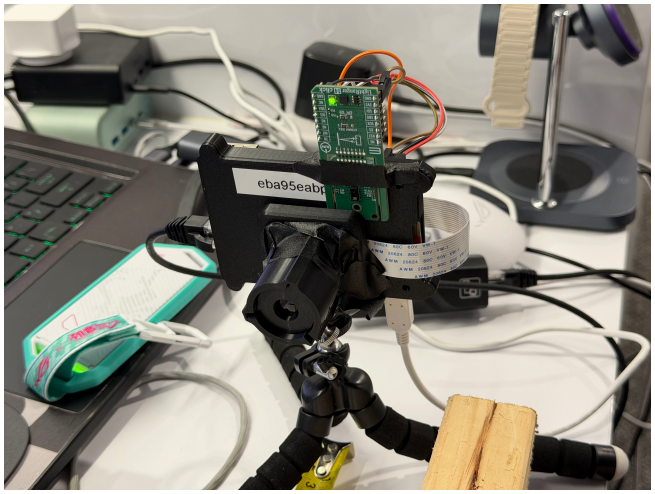


Fig. 1. Supplementary experiment apparatus. (a) Manual guide path and printed target. (b) Camera and depth board fixed to the kit plate. The team built the guide; the upright stand and kit plate were supplied. The photographs show placement; alignment and mounting stiffness were not measured.

We fixed focus, aperture, monitor position and brightness setting. The monitor was set to 160 Hz and brightness-menu value 90; luminance and optical transition times were not measured separately.

3.2 Acquisition and independent repeats

There are 40 runs and 1239 readable DNG images, including calibration data, stationary controls and 16 motion attempts. The motion attempts comprise 12 eligible independent pushes, two pilots and two weak-motion originals. The primary pushes cover smooth and variable instructions under coded and steady-white illumination, with three pushes per combination. In the smooth case, the operator attempts to make the push steady; in the variable case, the operator deliberately alters the speed. Measured velocity can vary under either instruction. The pilots and weak-motion originals remain outside the primary set; extra independent pushes replace the weak variable-motion attempts.

Each run stores the camera DNGs repeated at each exposure, and continuous 8×8 depth arrays. We record additional information for coded runs; these include the illumination events received by the Pi. For this study's primary set of analyses, we use a single exposure for each push for the analyses; multiple frames and zones from a push are not considered to be independent trials in the set. We request exposure and analogue gain at 1 s, and 1. The exposure is given in the metadata to be of 999942 μs; the gain was not independently measured. The acquisition is of 640×480 but has been downsampled to 240×180 linear-green images for the analyses.

The monitor repeats the binary code `1101001101` with nominal 200 ms states, a 2 s period and 60% white duty cycle. We play the code for 12 s within a 20 s recording. The monitor changes the paper's illumination while the camera integrates continuously; each one-second exposure uses its own logged state history for the model weights. White and coded runs use separate pushes, with unmatched trajectories and light doses.

3.3 Exposure selection

The selection rule was fixed prior to calculating reconstruction scores, and no preregistration was done. A primary exposure requires coverage of depth, solid coverage of coded light events, an approaching target and a minimum of 15 mm of travel during integration. We eliminate windows crossing an atypical event interval, then choose the greatest amount of travel in the case of a smooth push or the biggest departure from endpoints-to-endpoints linear depth in the case of a variable push. The two weak originals achieve maximum travel of just 9.05 and 6.99 mm and are scored individually. One of the coded runs has an event freeze after the selected time window. Its previous entirely covered window still can be used. Counting the number of events would not help uncover the issue.

4 Timing and Calibration

4.1 Clock convention and measurement density

We align the logs on the Pi monotonic clock. Its calendar was incorrect during recording, so original folder dates serve only as identifiers. We take the camera exposure window as $[t_s - T, t_s]$, interpreting t_s as the first-row readout/exposure-end timestamp and T as the recorded exposure. Rolling-row timing was not measured. Depth and screen logs use host receipt times; the saved device ticks and player clock were not calibrated to this timebase.

Each selected coded exposure contains 10–11 valid received depth frames in one second. We take the median of the central four zones (rows 4–5, columns 4–5, one-based), requiring at least three positive zones, depth 200–2000 mm and within-region spread no greater than 50 mm. We omit the known startup sample and interpolate between received frames. Raw depth is used without subtracting the 22 mm base-to-paper offset.

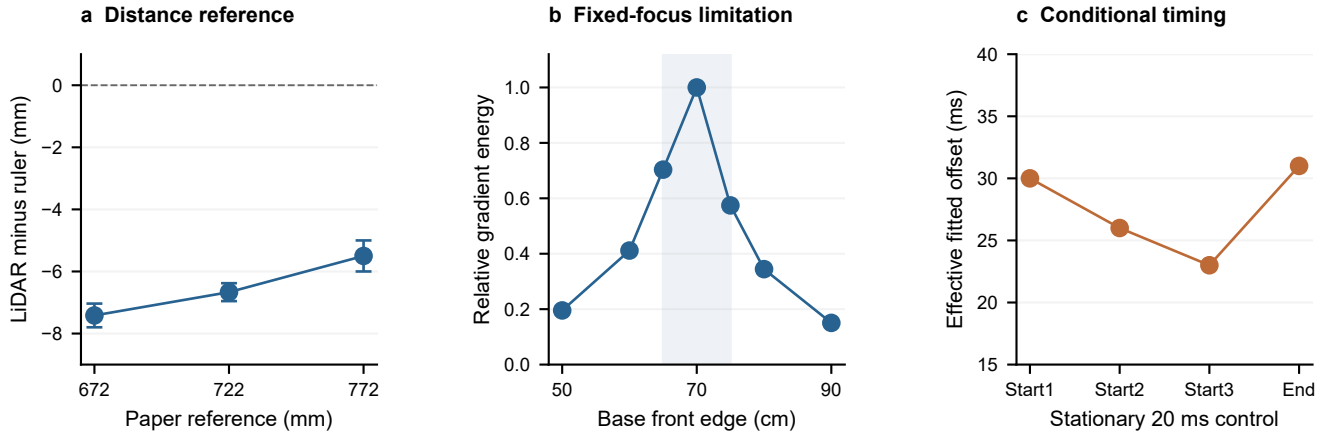


Fig. 2. Acquisition checks. (a) Depth residual against the paper reference, including the 22 mm base-to-paper offset. Bars show sample SDs of three technical run summaries; optical-origin uncertainty remains. (b) Registered gradient energy relative to 70 cm shows fixed-focus variation. It is a proxy of sharpness but it does not measure the resolution. The range 65–75 cm base that was kept for use is shaded. (c) Short-exposure controls give fitted offset (effective); the optical synchronization was not measured.

4.2 Stationary checks

At paper references 672, 722 and 772 mm, three technical run summaries give depths of 664.58 ± 0.38, 715.33 ± 0.29 and 766.50 ± 0.50 mm (mean $\pm$ sample SD). The residuals are -7.42, -6.67 and -5.50 mm. One near-labelled record matches the far position. It is retained as a position/reference discrepancy but excluded from this summary. Given that the origin in the optical data is not defined, the residual cannot be attributed completely to sensor errors.

Stationary 19.969 ms controls give fitted screen offsets of 30, 26 and 23 ms initially, and 31 ms at the end (end-fit $R^2 = 0.985$). We use 26 ms as an effective command-to-image offset. Each one-second control has only six covered observations, with weaker transferred-fit agreement ($R^2 = 0.814$, 0.607, 0.525). The short-exposure fit therefore leaves uncertainty in the long-exposure illumination history. The fixed-patch black/white raw-intensity ratio varies from about 0.167 to 0.211; we use 0.19 in the main model.

Relative gradient energy is 0.70 at 65 cm and 0.57 at 75 cm, compared with 70 cm. It falls to 0.20 and 0.15 at 50 and 90 cm (Fig. 2). These fixed-focus checks led us to restrict motion to the narrower range, where some defocus still remains. The selected motion images and final controls are unclipped; two initial white controls each contain one clipped raw pixel. The reference is the median of ten static 100 ms frames at the nominal 70 cm base position. These frames form one reference, not ten independent experimental repeats.

5 Reconstruction and Evaluation

5.1 Linear images and empirical axial geometry

We correct DNG values using per-channel black levels and white level, average the two green samples per Bayer cell, then resize by area averaging. The linear images for inversion are not JPEG-processed and do not apply any display gamma. SIFT feature matches and RANSAC similarity fits relate stationary early/late frames to the sharp reference; selected moving frames are excluded from endpoint fitting. The input and reference are separately globally normalized with the 3rd/97th-percentiles for contrast matching help. All reconstructions use the same input transform.

Endpoint scales calibrate a restricted axial model. With far/near depths z_f, z_n, scale ratio R, reference depth z_r and a fitted fixed pivot,

$$\beta(z) = \frac{z^{-1} - z_f^{-1}}{z_n^{-1} - z_f^{-1}}, \qquad s(z) = R^{\beta(z)-\beta(z_r)}. \tag{1}$$

This interpolates log scale over inverse depth using the measured endpoints, without requiring an exact optical-origin distance from the ruler. The main dynamic warp scales about the pivot. Endpoint rotation is used for reference registration and enters the dynamic warp only in a sensitivity check. Range alone does not recover continuous target tilt or lateral drift.

5.2 LiDAR-guided axial scale-blur model

An exposure combines differently scaled views of one sharp reference-plane image x. We write the observation y as

$$\begin{aligned} y &= A(z,u)x + \epsilon, \\ A &= \sum_{k=1}^{48} w_k W[s(z_k)], \\ w_k &= \frac{0.19 + 0.81u_k}{\sum_j (0.19 + 0.81u_j)}, \end{aligned} \tag{2}$$

where z_k is interpolated depth, $W[s(z_k)]$ scales about the fitted pivot, and u_k is the logged binary state after the timing shift. The 48 midpoint bins approximate the exposure integral from 10–11 received depth frames. W is implemented via sparse bilinear interpolation with replicated boundaries. The weights are an approximation of the relative irradiance

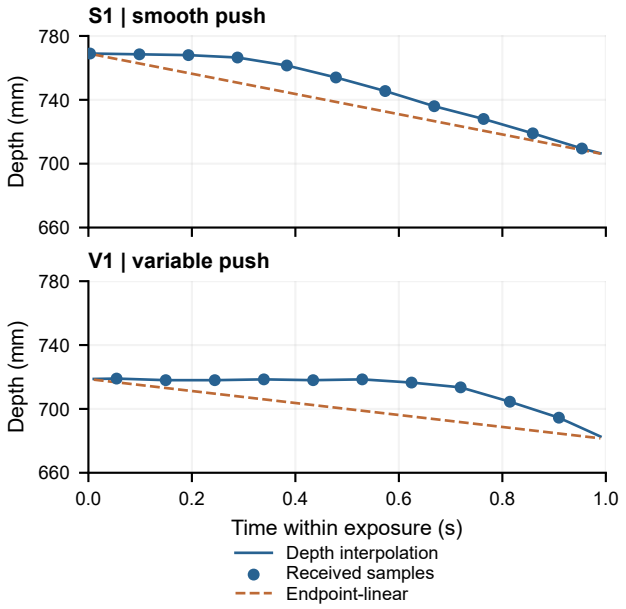


Fig. 3. Depth during two selected exposures. S1 and V1 contain 11 and 10 valid received samples. The models share the same exposure-boundary depths and differ only in the interior trajectory. A smooth-push instruction does not impose constant speed.

with 0.19 floor on the illumination in the black state. Each of the columns of A is the response to one point in the image - a discrete point spread function (PSF). Expansion moves points farther from the fixed scaling pivot more strongly, so the PSF varies spatially across the image. Reconstruction therefore requires inverting a spatially varying operator.

5.3 TV-regularised reconstruction

Once the records and endpoint geometry fix A, we estimate the image by non-blind, TV-regularised least squares:

$$\hat{x} = \arg\min_{0\le x\le 1} \tfrac{1}{2}\|Ax - y\|_2^2 + \lambda\,\mathrm{TV}(x), \qquad \lambda = 0.002. \quad (3)$$

The data term fits the recorded blur, while TV limits unstable fluctuations. We use 320 outer iterations of an accelerated proximal-gradient scheme inspired by FISTA [3], a power-estimated step size and up to 35 inner TV iterations. The TV step is approximate, and clipping constrains intensities to $[0, 1]$. The comparator replaces measured $z(t)$ with constant depth velocity between the same boundary depths. Both methods share the image, light weights, geometry, solver and mask. We also evaluate the blurred input and denoise-only processing ($A = I$). Parameters are shared across trials.

5.4 Scoring and numerical checks

We map all 16 attempts into one canonical reference frame and fix a geometry-only intersection mask before scoring (Fig. 4). PSNR uses unit intensity range; Gaussian-window SSIM uses $\sigma = 1.5$ and population covariance [5]. Methods receive no separate post-reconstruction alignment or contrast fit. The reference is used for registration, normalization and scoring, making this a reference-assisted experiment.

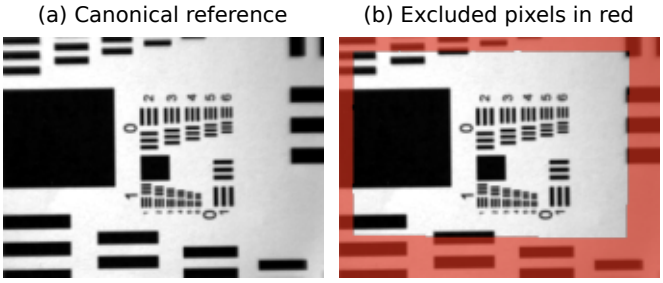


Fig. 4. A common reference and common mask for all trials and all methods. Excluded pixels marked in red. Scores use the remaining 28327 pixels (65.6% of the 240×180 image) within the irregular mask, rather than only its boundary or a rectangular display crop.

TABLE 1
Six independent coded exposures scored on the same canonical mask. I: blurred input; C: endpoint-linear; L: LiDAR. S/V denote smooth/variable push instructions, with independent repeats numbered 1–3. Real speed may change in both cases.

	PSNR (dB)			SSIM		
Trial	I	C	L	I	C	L
S1	15.13	18.08	18.21	0.646	0.728	0.767
S2	13.57	19.15	18.04	0.564	0.789	0.765
S3	16.40	14.33	18.38	0.712	0.664	0.777
V1	18.71	14.29	19.22	0.791	0.647	0.826
V2	11.59	16.21	25.09	0.523	0.745	0.906
V3	21.45	17.70	19.07	0.878	0.803	0.840

Identity, row-sum and adjoint errors are below 8×10^{-7}. Synthetic inversion improves PSNR from 13.99 to 27.15 dB. Doubling the iterations from 320 to 640 changes PSNR by at most 0.049 dB; changing the bins from 48 to 96 changes it by at most 0.207 dB. These checks assess numerical integrity. The physical captures test how well the model describes practice.

6 Results and Analysis

6.1 Independent coded pushes

Table 1 includes every primary coded push. Targets travel 27.8–63.4 mm during the selected exposure within the nominal 100 mm path. The measured trajectories in Fig. 3 show motion that the endpoint-linear model misses.

Measured depth beats endpoint-linear in five of six trials on both metrics. Mean paired differences are $+3.038 \pm 3.666$ dB and $+0.0843 \pm 0.0797$, respectively (sample SD across six independent pushes). Figure 6 shows each paired difference and the mean with one sample SD. These are reported as descriptive statistics. Against the blurred input, mean gains are $+3.523 \pm 5.421$ dB and $+0.1280 \pm 0.1485$, also with five positive trials. The median paired PSNR gain is 2.706 dB. Leaving out each push in turn gives mean gains of 1.870–3.869 dB; omitting V2, the largest gain, gives 1.870 dB. The primary result retains all six trials; these omissions only check how much each push affects the mean.

S2 loses 1.115 dB and 0.0239 SSIM to the endpoint-linear model, although both reconstructions beat its input. V3 improves over endpoint-linear but loses 2.383 dB and 0.0380 SSIM to input. V2 has the largest paired PSNR gain,

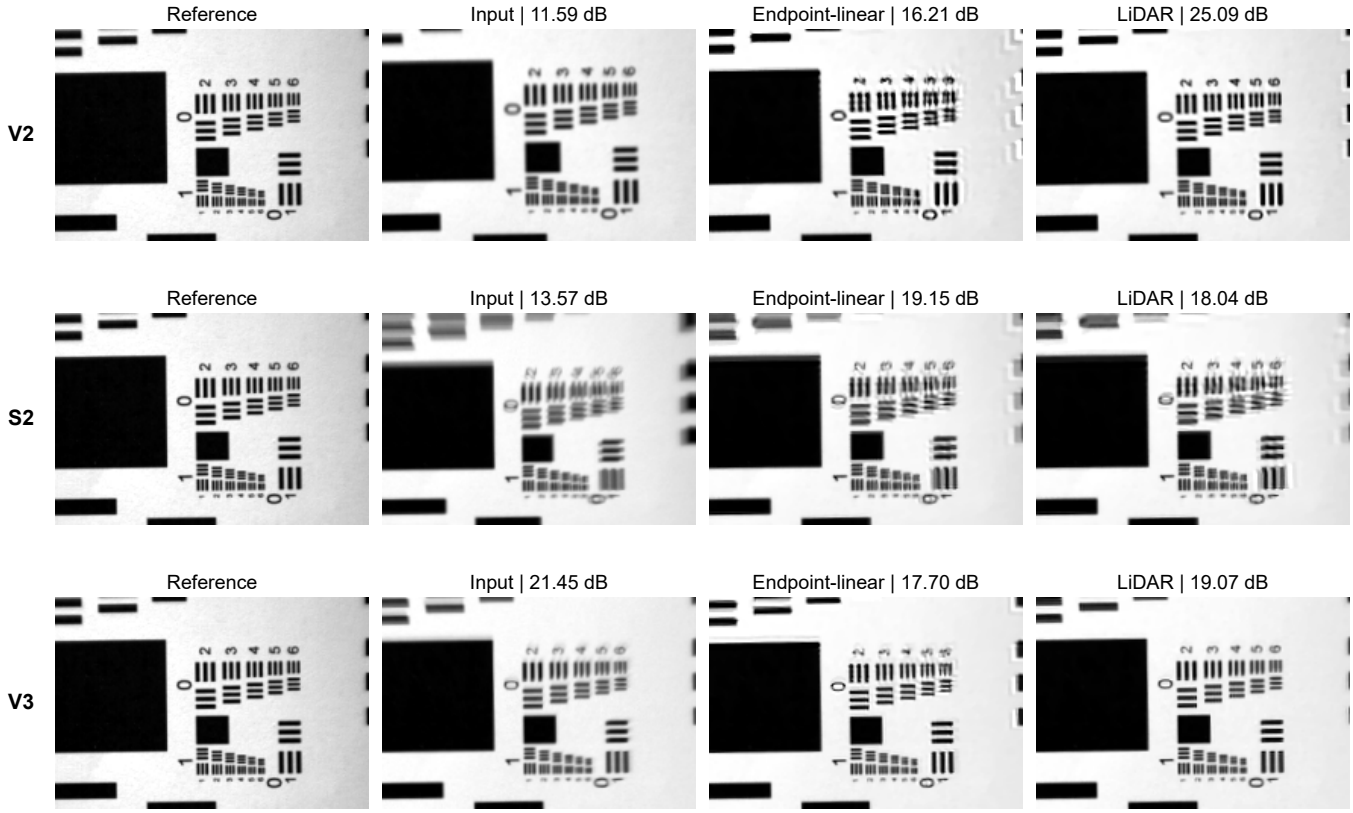


Fig. 5. Three outcomes using the same canonical crop and linear grayscale range $[0, 1]$. V2 has the largest LiDAR advantage over endpoint-linear; S2 favours endpoint-linear; V3 favours the input over either reconstruction. All six trials are in Table 1 and the full analysis gallery. Scores use the irregular common mask within each display crop.

8.880 dB. Figure 5 shows V2 and both failure types. These represent two distinct failures, one where the measured depth loses out against the simpler motion model and the other where the motion inversion is degrading the already superior input.

6.2 Examining the failed comparisons

In all 13 one-factor settings, S2 remains under endpoint-linear, with pairs of PSNR scores differing by between -1.209 and -0.847 dB. Altering any of the checked timing, TV weight, light floor level, depth offset, or rotation angle does not reverse the result. Its physical cause remains unresolved.

For V3, a denoise-only process yields 21.43 dB, close to the original 21.45 dB, whereas the LiDAR inversion results in a score of 19.07 dB. Most of the loss occurs after motion inversion. Increasing the solver from 320 to 640 iterations leaves the score at 19.07 dB. With 96 integration bins, the score is 19.10 dB. TV weights spanning 0.0005 to 0.02 lead to a score of 19.04–19.12 dB. The loss remains even after all these checks. Equation (2) fits the recorded blur, so errors in the assumed illumination or motion may lead the solver away from the static reference. Pose, defocus, lighting and timing remain possible contributors; these tests do not isolate their effects.

TABLE 2
All primary cells, $n = 3$ independent pushes each. Δ_{LC} is LiDAR minus endpoint-linear PSNR (mean $\pm$ sample SD); Δ_{LI} is LiDAR minus input, also as mean $\pm$ sample SD. Per-trial scores are supplied.

Light	Instruction	Δ_{LC} (dB)	Δ_{LI} (dB)
Coded	Smooth	1.020 ± 2.694	3.174 ± 1.248
Coded	Variable	5.057 ± 3.758	3.871 ± 8.458
White	Smooth	1.549 ± 2.370	3.422 ± 2.840
White	Variable	1.396 ± 2.118	-0.932 ± 0.898

6.3 Steady-white controls and repeat variation

Table 2 summarizes the 12 main pushes, with three independent repeats per cell. Mean model advantage was higher for the variable-coded than the smooth-coded cell. White-variable reconstruction is better on average than endpoint-linear, but not better than the input. Motion and integrated illumination differ between white and coded pushes, preventing a causal comparison of coding.

6.4 Stationary and sensitivity controls

Relative to denoise-only processing, measured-depth reconstruction gains 3.522 ± 5.411 dB on average (sample SD, $n = 6$), with five trials improving in both metrics. Thus TV denoising alone does not account for the observed mean gain.

Three static depth-driven controls change PSNR by only $+0.001$, $+0.026$ and $+0.010$ dB relative to input, with SSIM

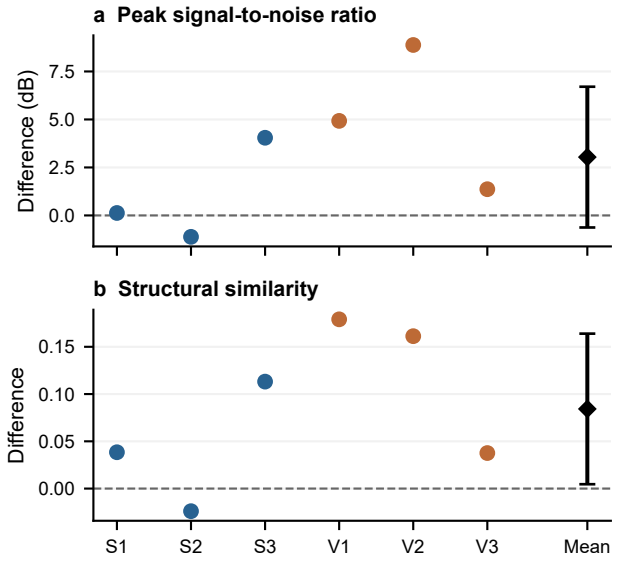


Fig. 6. LiDAR-minus-endpoint-linear differences for all six coded pictures. The black diamond shows the mean and one sample SD; it is separate from the six trials and does not represent a confidence interval. Blue: smooth instructions; orange: variable instructions.

changes -0.0029, -0.0028 and -0.0022. The small changes show how the pipeline behaves without target motion; they provide a control for the moving trials.

We additionally experiment with 13 one-factor variations on TV weight, light offset, depth timing offset (± 50 ms), black-state floor, common depth offset (± 22 mm) and endpoint rotation. The average coded LiDAR benefit is 2.43–3.56 dB and the individual trial signs may change. This interval describes the settings which are tested individually; it does not provide a statistical bound on the uncertainty nor a joint worst-case figure. Doubling the quadrature bins changes the mean benefit from 3.038 to 3.013 dB. Frozen-geometry replay in a compatible numerical setting changes PSNR by at most 0.0032 dB, illustrating the repeatability of the frozen-geometry calculation. The subpixel fitting might be affected by fresh registration.

7 Limitations and Pitfalls

Command receipt times and a fitted offset are used to estimate the illumination history. We neither measured optical transitions with a photodetector, nor calibrated the ranging-to-receipt delay or measured the rolling-row skew. The timing perturbations we test represent only some of the possible errors.

We also left out from the scale model continuous tilt and lateral drift, depth-dependent defocus and spatial variation in illumination. We observe a change in sharpness even inside the range retained in our tests of fixed-focus control. The optical origin remains uncertain, and the 22 mm base-to-paper offset does not provide an independent sensor-bias calibration. These effects are not separated experimentally and we are not able to link either failed comparison to one physical cause.

Reference-assisted registration and contrast normalization limit the scope of the scores. Downsampling and TV may suppress fine bars, and the shared mask depends on reference geometry. Selection was exploratory; all attempts and weak-motion replacements are documented. Six coded pushes and three repeats per cell support descriptive comparisons. Unmatched white/coded motion and light dose prevent a causal claim about coding.

8 Conclusions and Future Work

Our manual-push system records long-exposure images alongside depth and monitor-command histories for axial-motion reconstruction. For six independent coded exposures, LiDAR-guided reconstruction gives a mean PSNR improvement of 3.04 dB over a matched endpoint-linear model. Five trials improve both PSNR and SSIM. S2 lost to that endpoint-linear model, while V3 lost to the blurred input image. The control and numerical verification indicated a conditional gain from measured motion; timing, focus, pose and reference assistance still limit the result.

Given two additional months, we would first measure the optical transition and the rolling row timing, as well as the range origin and the image scale and illumination on the path. Then, we would preregister the selection rules and collect randomized repeated motions, at equivalent light dose. A continuous pose measurement and a defocus model dependent on depth would help disentangle possible causes of failure, before testing the optimal codes.

9 Reproducibility

The accompanying package includes the acquisition data, all 16 attempt scores, frozen selections and geometry, source hashes, reconstruction arrays, scripts, dependencies and build instructions. Run `python reproduce.py` to recalculate saved-array metrics and reproduce the paired-effect and mask figures. Reproduce all 16 RAW inversions with frozen geometry using `python reproduce.py --reconstruct`, or one coded exposure using `python reproduce.py --trial S1`. The README records the tested environment and hardware; fresh registration can vary numerically. The previous five-trial dataset is separate. Code and RAW data: https://github.com/Jarvis-xuhui-du/MECH5720_Major_project/releases/tag/raw-data-v1.0.

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